Book XI of Euclid's Elements

Formal Reconstruction — Technical Documentation

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Chapter 4

Proposition XI.4

4.1 Statement

Euclid XI.4 gives the basic criterion for perpendicularity to a plane: if a line through the common point of two intersecting lines of a plane is perpendicular to each of those two lines, then it is perpendicular to the plane itself.

The formal reconstruction is implemented in

```
CGJteamLab/Proposition11_4.lean
```

on top of the spatial infrastructure in

```
CGJteamLab/Hilbert3DAxioms.lean
CGJteamLab/Hilbert3DInterface.lean
```

The Lean source is authoritative for theorem signatures, helper names, actual dependencies, and proof status.

4.2 Euclid XI.3 and the target notion

Book XI, Definition 3 says that a straight line is perpendicular to a plane when it makes right angles with every straight line which meets it and lies in that plane. The project encodes exactly this universal synthetic notion.

For two incidence lines:

```
def HilbertLinesPerpendicularAt
    [H : HilbertIncidence Geo]
    (l m : Geo.Line)
    (O : Geo.Point) : Prop :=
  H.OnLine O l /\
  H.OnLine O m /\
  exists A B : Geo.Point,
    Ne A O /\
    Ne B O /\
```

```
    H.OnLine A l /\
    H.OnLine B m /\
    Not (PrimCollinear Geo A O B) /\
    HilbertRightAngle Geo A O B
```

The explicit noncollinearity condition is important. It prevents a degenerate "self-perpendicular" witness and yields the reusable theorem

```
hilbert_linesPerpendicularAt_ne
```

showing that perpendicular lines are distinct.

For a line and a plane:

```
def HilbertLinePerpendicularPlaneAt
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    (l : Geo.Line)
    (pi : S.Plane)
    (O : Geo.Point) : Prop :=
  H.OnLine O l /\
  S.OnPlane O pi /\
  forall m : Geo.Line,
    HilbertLineInPlane Geo m pi ->
    H.OnLine O m ->
    HilbertLinesPerpendicularAt Geo l m O
```

Thus the conclusion of XI.4 is not a metric abbreviation. It is literally the universal quantification required by Euclid XI.Def.3.

4.3 Classical configuration

Let m, n be two distinct straight lines of a plane π, meeting at O, and let l be a line through O perpendicular to both.

The desired conclusion is that l is perpendicular to every line $q \subset \pi$ through O.

4.4 Euclid's classical proof

Euclid makes the argument constructive.

Take equal segments on the two given plane lines:

$$AE = EB = CE = ED,$$

where E is the common point of the two given lines. Draw an arbitrary straight line GEH through E in the plane, choose an arbitrary point F on the elevated perpendicular, and join the auxiliary segments.

The classical dependency chain is:

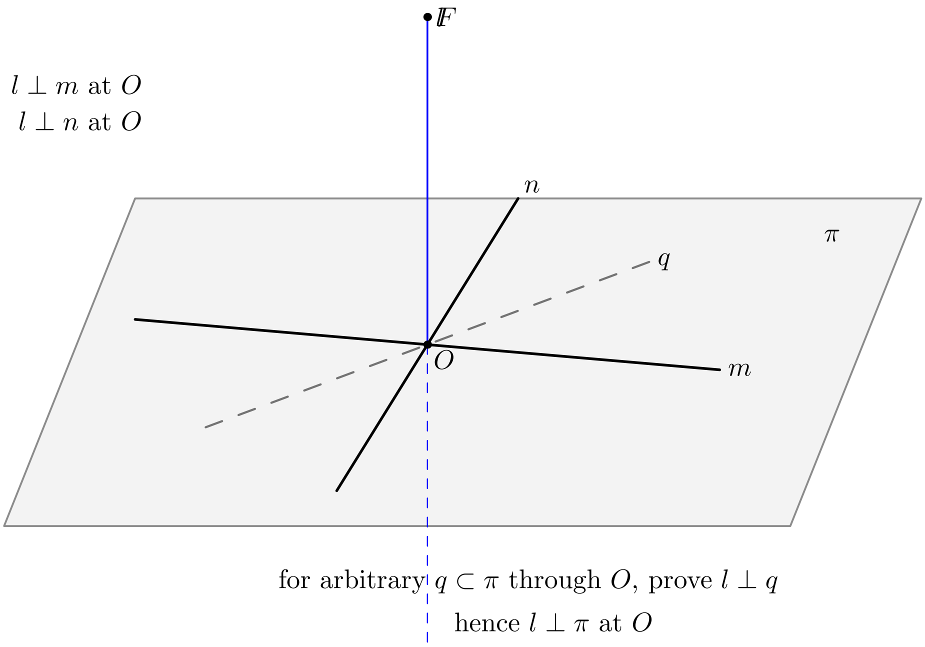

Figure 4.1: Classical configuration for Euclid XI.4. The ambient line l passes through O and is perpendicular at O to two distinct lines m, n lying in the plane π. The dashed line q represents an arbitrary line of π through O, which is the universal target of the conclusion.

1. Vertical angles at E are equal (I.15). Together with the four equal radial segments, I.4 gives
$$AD = CB$$
and a corresponding angle equality.

2. Applying I.26 to the planar transversal gives
$$GE = EH, \qquad AG = BH.$$

3. Since FE is perpendicular to both original plane lines, I.4 applied to two right triangles gives
$$FA = FB, \qquad FC = FD.$$

4. I.8 compares two spatial triangles and supplies the angle relation needed to transport the planar data to the arbitrary transversal.

5. A further I.4 gives
$$FG = FH.$$

6. Finally, from
$$GE = EH, \qquad FE = FE, \qquad FG = FH,$$
I.8 yields
$$\angle GEF = \angle HEF.$$
Since $G - E - H$, the adjacent equal angles are right angles.

7. The transversal GEH was arbitrary. Hence FE makes a right angle with every line through E in the plane. By XI.Def.3, $FE \perp \pi$.

This is a genuinely three-dimensional use of Book I congruence theory: several compared triangles are not assumed to lie in one common plane.

4.5 Formal theorem

The production theorem is:

```
theorem euclid_proposition_11_4
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (pi : S.Plane)
    (m n : PlaneLine Geo pi)
    (l : Geo.Line)
    (O : PlanePoint Geo pi)
    (hmn : Ne m n)
    (hPerpM :
      HilbertLinesPerpendicularAt Geo l m.1 O.1)
    (hPerpN :
      HilbertLinesPerpendicularAt Geo l n.1 O.1) :
    HilbertLinePerpendicularPlaneAt Geo l pi O.1
```

The theorem does not assume separately that $l \neq m$ or $l \neq n$. Those facts follow from the strengthened nondegenerate line–line perpendicularity predicate.

4.6 Spatial architecture

4.6.1 No ambient planar type-class instance

The ambient geometry is three-dimensional. The proof deliberately does *not* install a global instance

```
HilbertCongruence Geo
```

or

```
HilbertOrder Geo
```

on the whole space. Planar reasoning is performed only in explicit plane slices

$$\mathrm{PlaneGeo}(\pi).$$

The corresponding points and lines are subtypes:

```
PlanePoint Geo pi
PlaneLine  Geo pi
```

and the interface provides bridges between induced planar facts and ambient facts, for example

```
planeGeo_congruent
planeGeo_angleCongruent_iff_ambient
planeGeo_sameRay_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_linesPerpendicularAt_iff_ambient
```

This separation is one of the central architectural features of the proof.

4.6.2 Spatial congruence

Euclid freely applies I.4 and I.8 to triangles in different planes. The project reconstructs this through the spatial congruence layer rather than by coordinates or by an ambient Euclidean metric.

The general spatial infrastructure includes:

```
hilbert_space_congruent_reflexive
hilbert_space_congruent_symmetry
hilbert_space_sas_remaining_angles
hilbert_space_sas_third_side_and_angle
hilbert_space_sss_angleA_in_plane
```

The target triangle may be placed in its own explicit plane and compared with an ambient source triangle. This is the formal counterpart of Euclid's unqualified use of plane triangle congruence inside solid geometry.

4.7 The balanced cross

The first construction is the formal analogue of Euclid's

$$AE = EB = CE = ED.$$

The helper

```
planeGeo_opposite_points_on_line_congruent_to
```

constructs opposite equal points on one line. Applying it twice yields

```
hilbert_XI4_balanced_cross_exists
```

and points A, B, C, D satisfying

$$A - O - B, \qquad C - O - D$$

and

$$OA \cong OR, \quad OB \cong OR, \quad OC \cong OR, \quad OD \cong OR.$$

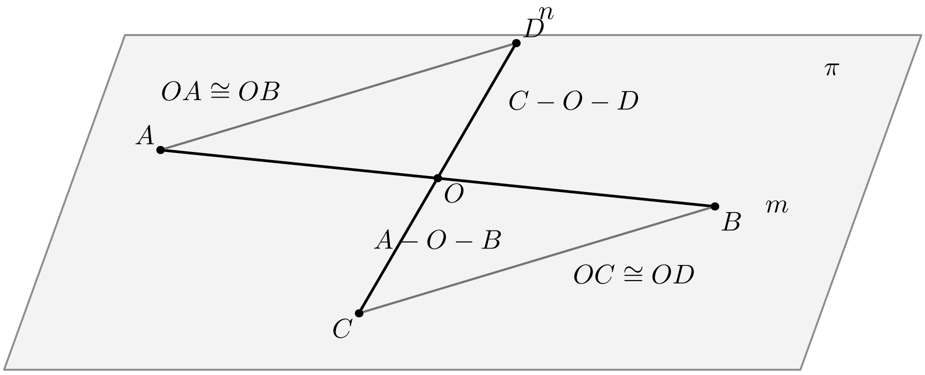

Figure 4.2: Balanced cross in π. Points A, B lie on m, points C, D lie on n, with $A - O - B$, $C - O - D$, $OA \cong OB$, and $OC \cong OD$. The connector segments AD and BC are the first pair used by Euclid's congruence argument.

The first triangle comparison is isolated by

```
planeGeo_XI4_cross_noncollinear
hilbert_XI4_balanced_cross_first_SAS
```

and proves the ambient consequences

$$AD \cong BC, \qquad \angle OAD \cong \angle OBC.$$

This is the formal Hilbert version of Euclid's first use of I.4.

4.8 Fixed-transversal prototype

Before the proof was generalized to every transversal, a fixed connector configuration was developed. Its role is mathematically transparent: given

$$A - G - D, \qquad B - J - C, \qquad G - O - J,$$

the balanced cross implies

$$OG \cong OJ, \qquad AG \cong BJ.$$

The relevant planar ingredients are

```
planeGeo_XI4_transversal_noncollinear
hilbert_XI4_transversal_ASA
```

corresponding to Euclid's I.26 step.

The spatial point $F \in l$, together with the two original perpendicularities, then gives

$$AF \cong BF, \qquad CF \cong DF.$$

The proof packages this in

```
hilbert_XI4_spatial_perpendicular_bisector_equidistant
hilbert_XI4_spatial_opposite_pairs_equidistant
```

The spatial SSS/SAS chain is then:

```
hilbert_XI4_spatial_SSS_angle
hilbert_XI4_space_angle_sameRay_first_congruent
hilbert_XI4_spatial_SSS_angle_normalized
hilbert_XI4_spatial_SAS_transversal_side
hilbert_XI4_spatial_transversal_F_equidistant
hilbert_XI4_second_spatial_SSS_angle
hilbert_XI4_right_angle_of_adjacent_equal
```

Its mathematical content is

$$\begin{gathered} AD \cong BC, \quad AF \cong BF, \quad CF \cong DF \\ \Downarrow \\ \angle DAF \cong \angle CBF \\ \Downarrow \\ A - G - D,\ B - J - C \\ \Downarrow \\ \angle GAF \cong \angle JBF \\ \Downarrow \\ AG \cong BJ,\ AF \cong BF \\ \Downarrow \\ GF \cong JF \\ \Downarrow \\ OG \cong OJ,\ OF \cong OF,\ GF \cong JF \\ \Downarrow \\ \angle GOF \cong \angle JOF \\ \Downarrow \\ G - O - J \\ \Downarrow \\ \mathrm{RightAngle}(G, O, F). \end{gathered}$$

The last implication is exactly Euclid's definition of a right angle as two adjacent equal angles formed by a straight line.

4.9 The arbitrary transversal: the real formal difficulty

The public conclusion quantifies over *every* line of π through O. A conventional drawing suggests that such a transversal meets the two connector segments AD and BC, but that is false in general. This was the key positional issue in the formal proof.

The correct argument uses two possible connector families:

$$AD/BC \qquad \text{or} \qquad BD/AC.$$

For a transversal $q \neq m, n$, apply Pasch to the triangle ABD. The line q enters through AB at O, hence it exits either through AD or through BD.

This is isolated as

```
hilbert_XI4_transversal_pasch_first_exit
```

Next,

```
hilbert_XI4_opposite_point_on_transversal
```

constructs the point J opposite G on q with

$$G - O - J, \qquad OG \cong OJ.$$

Two neutral landing lemmas then prove:

```
hilbert_XI4_first_exit_opposite_lands_on_BC
hilbert_XI4_second_exit_opposite_lands_on_AC
```

so that the opposite point lands on the correct paired connector. The combined package is

```
hilbert_XI4_arbitrary_transversal_connector_package
```

and returns

$$G, J \in q, \quad G - O - J, \quad OG \cong OJ,$$
$$\text{and either}$$
$$A - G - D,\ B - J - C,$$
$$\text{or}$$
$$B - G - D,\ A - J - C.$$

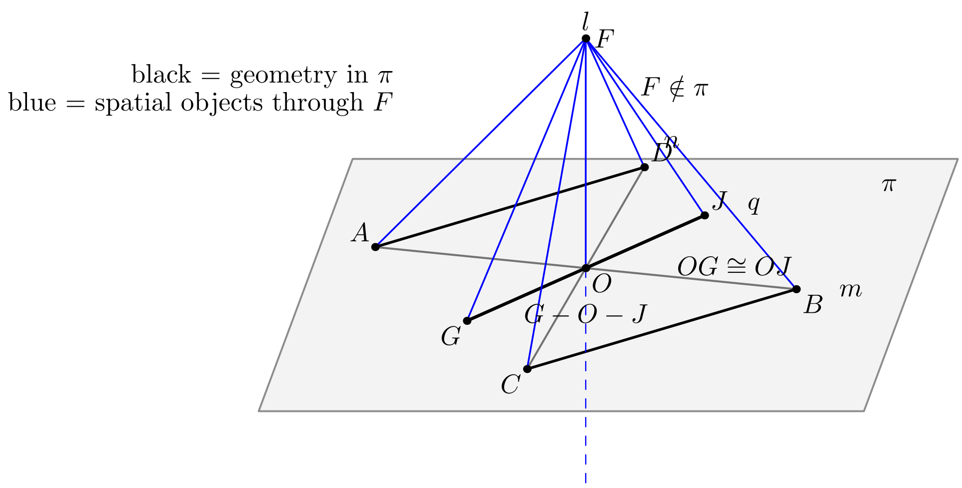

Figure 4.3: Arbitrary-transversal stage. For $q \subset \pi$ through O, Pasch selects one of the two connector families. The opposite point J is constructed with $G - O - J$ and $OG \cong OJ$. The spatial SSS/SAS chain then proves that $\angle GOF$ is right.

This entire stage uses incidence, order, Pasch, and congruence. It does not use the Euclidean parallel axiom.

4.10 Right angle on every transversal

With the connector package available, both branches reduce to the same spatial SSS/SAS scheme. The theorem

```
hilbert_XI4_arbitrary_transversal_right_angle
```

states, in substance:

$$q \subset \pi, \quad O \in q, \quad q \neq m, n,$$
$$F \notin \pi, \quad AF \cong BF, \quad CF \cong DF$$

imply the existence of $G \in q$, $G \neq O$, such that

$$\mathrm{RightAngle}(G, O, F).$$

The two exceptional cases $q = m$ and $q = n$ are discharged directly from the hypotheses of XI.4.

4.11 Why the point F is outside the plane

The core proof chooses an arbitrary nonvertex point F on l. To use spatial noncollinearity cleanly, it proves

$$F \notin \pi.$$

This is the theorem

```
hilbert_XI4_perpendicular_point_off_plane
```

The proof is neutral and conceptually important.

Assume $F \in \pi$. Then the line l, determined by O and F, is itself a line of π. Transfer the two ambient perpendicularities to `PlaneGeo pi`. Choose nonvertex points $A \in m$ and $C \in n$. Then both

$$\angle FOA \quad \text{and} \quad \angle FOC$$

are right.

The helper

```
hilbert_XI4_two_right_angles_same_first_arm_collinear
```

uses Hilbert III.4 uniqueness to prove that A, O, C are collinear. Line uniqueness then forces $m = n$, contradicting the hypothesis.

No parallel postulate is used.

4.12 Closing the universal line–plane statement

The core theorem

```
hilbert_XI4_line_perpendicular_plane_core
```

does the following:

1. choose $F \neq O$ on l;
2. construct the balanced cross A, B, C, D in π;
3. prove $F \notin \pi$;
4. prove $AF \cong BF$ and $CF \cong DF$;
5. let q be an arbitrary line of π through O;
6. if $q = m$ or $q = n$, use the hypotheses;
7. otherwise invoke the arbitrary-transversal theorem and obtain a point $G \in q$ such that $\angle GOF$ is right;
8. construct the witness required by `HilbertLinesPerpendicularAt Geo l q O`.

There is one subtle final representation step. The right angle obtained from the transversal theorem is oriented as

$$\angle GOF.$$

The line-line perpendicularity witness is more convenient with the order

$$\angle FOG.$$

The proof does not use an ambient planar congruence instance to swap the arms. Since l and q intersect at O, they determine an explicit plane ρ. Inside `PlaneGeo rho` the right angle is swapped using the ordinary planar congruence API, and the result is transported back to the ambient geometry.

This preserves the 3D architecture: all planar congruence reasoning occurs inside an explicit plane slice.

4.13 Complete formal DAG

The proof architecture can be summarized as follows.

```
two distinct plane lines m,n through O
+ l perpendicular to m and n at O
-> balanced cross A-O-B, C-O-D
-> AD  = BC and angle OAD  = angle OBC
-> choose F != O on l
-> F notin pi
-> AF  = BF, CF  = DF
-> arbitrary q in pi through O
-> if q=m or q=n:  hypothesis
-> otherwise Pasch in triangle ABD
-> exit through AD or BD
-> opposite J on q, G-O-J, OG  = OJ
-> landing on BC or AC
-> ASA: AG  = BJ (or BG  = AJ)
-> spatial SSS/SAS: GF  = JF
-> spatial SSS: angle GOF  = angle JOF
-> G-O-J -> RightAngle GOF
-> l perpendicular q at O
-> forall q in pi through O, l perpendicular q
-> HilbertLinePerpendicularPlaneAt Geo l pi O.
```

4.14 Classical dependencies versus Lean dependencies

Role	Euclid	Lean reconstruction
Four equal radial segments	cut off AE, EB, CE, ED equal	opposite-point construction on each of the two lines inside `PlaneGeo pi`
Vertical angles	I.15	developed vertical-angle and angle-congruence infrastructure
First triangle congruence	I.4	balanced-cross SAS helper
Transversal equalities	I.26	Pasch, transversal ASA, and the two landing branches
Distances from F	I.4 on right triangles	spatial perpendicular-bisector/equidistance helper
Cross-plane triangle comparison	I.8 and I.4	general spatial SSS/SAS API through explicit carrier planes
Arbitrary line in the plane	“drawn at random”	universal q; cases $q = m$, $q = n$, or the Pasch branch
Line perpendicular to plane	XI.Def.3	the universal line–plane perpendicularity predicate

The Lean DAG is longer because it must prove all incidence, order, noncollinearity, carrier-plane, and orientation facts that the classical diagram leaves implicit. The geometric argument itself remains Euclidean XI.4.

4.15 Axiomatic status

The public theorem assumes the spatial Hilbert hierarchy

```
HilbertIncidence Geo
HilbertPlaneIncidence Geo
HilbertSpacePrimitive Geo
HilbertSpaceIncidence Geo
```

```
HilbertSpaceOrder
HilbertSpaceCongruence
```

No `HilbertEuclideanPlane` instance occurs in the theorem signature. In particular, the proof of XI.4 does not use Hilbert Group IV, the Euclidean parallel axiom.

No continuity or Archimedean axiom is introduced by XI.4.

The final Lean audit is:

```
'Geometry.euclid_proposition_11_4' depends on axioms:
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The project deliberately retains the inherited `bookZero_nullSegment1` dependency at this stage. XI.4 introduces no new proposition-specific axiom.

4.16 Formalization notes

4.16.1 The proof is synthetic

There are no coordinates, vectors, scalar products, normal vectors, trigonometric functions, angle measures, or real-valued lengths.

All claims are expressed through:

- incidence of points, lines, and planes;
- strict betweenness;
- same-ray and side-of-line relations;
- segment and angle congruence;
- Hilbert right angles;
- Pasch;
- planar and spatial SAS/SSS/ASA arguments.

4.16.2 The point F is not a normal-vector surrogate

The point F is merely a nonvertex point chosen on the candidate line l. It is used to form ordinary synthetic triangles such as AOF, BOF, GOF, and JOF. The proof never assigns coordinates to F and never interprets OF as a vector.

4.16.3 Why `PlaneGeo` is essential

Some arguments are intrinsically planar: Pasch, same-side uniqueness of an angle copy, same-ray normalization, and ordinary Hilbert angle construction. Other arguments compare triangles in different planes. The proof therefore alternates explicitly between:

$$\text{ambient 3D geometry} \quad \longleftrightarrow \quad \text{PlaneGeo}(\pi) \quad \text{or} \quad \text{PlaneGeo}(\rho).$$

This is not implementation noise. It records exactly where Euclid is using plane geometry inside a solid-geometric proof.

4.16.4 Hidden positional data

The formal proof makes explicit all of the following facts:

- the two given plane lines are genuinely distinct;
- perpendicularity witnesses are nondegenerate;
- the four balanced points lie on the intended carriers;
- $A - O - B$ and $C - O - D$ are strict orders;
- the auxiliary point F is outside π;
- a general transversal need not meet the fixed pair AD, BC;
- Pasch selects one of two connector families;
- the opposite point on the transversal lies on the paired connector;
- every spatial triangle used by SSS or SAS is noncollinear;
- the final arm swap is performed in the explicit plane through l and the arbitrary transversal.

These facts are precisely the information supplied visually, but not logically, by a conventional diagram.

4.16.5 Why the two connector families matter

A tempting but incorrect formalization is to assume that every line through O meets both segments AD and BC. The diagram can make this appear plausible, but it is not true.

The final proof instead analyzes the first exit of the transversal from triangle ABD. This produces exactly the dichotomy required by the geometry:

$$(AD, BC) \quad \text{or} \quad (BD, AC).$$

This is one of the most significant proof-design corrections in the XI.4 development.

4.17 Reusable infrastructure produced by XI.4

The proposition required one general correction to the 3D interface: line-line perpendicularity now carries an explicit noncollinearity witness, so that distinctness of the carriers is a theorem rather than an external hypothesis.

The general 3D interface also exposes the bridges

```
HilbertLinesPerpendicularAt
hilbert_linesPerpendicularAt_ne
planeGeo_linesPerpendicularAt_iff_ambient
HilbertLinePerpendicularPlaneAt
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
HilbertLinePerpendicularPlaneAt.incidence
```

The remaining declarations beginning with `hilbert_XI4_` are kept in `Proposition11_4.lean` because they package the concrete Euclidean XI.4 configuration.

4.18 Dependency summary

$$\begin{gathered}
\text{Hilbert spatial incidence + order + congruence} \\
\Downarrow \\
\text{PlaneGeo slices and ambient/plane bridges} \\
\Downarrow \\
\text{nondegenerate line-line perpendicularity} \\
\Downarrow \\
\text{balanced cross } A - O - B,\ C - O - D \\
\Downarrow \\
AD \cong BC,\ \angle OAD \cong \angle OBC \\
\Downarrow \\
F \in l,\ F \neq O,\ F \notin \pi \\
\Downarrow \\
AF \cong BF,\ CF \cong DF \\
\Downarrow \\
\text{Pasch for arbitrary } q \subset \pi \text{ through } O \\
\Downarrow \\
\text{two connector branches + opposite point } J \\
\Downarrow \\
GF \cong JF, \quad \angle GOF \cong \angle JOF \\
\Downarrow \\
\text{RightAngle}(G, O, F) \\
\Downarrow \\
l \perp q \text{ for every } q \subset \pi \text{ through } O \\
\Downarrow \\
\texttt{euclid_proposition_11_4}.
\end{gathered}$$

4.19 Status

Production theorem	`Geometry.euclid_proposition_11_4`
Production file	`CGJteamLab/Proposition11_4.lean`
General 3D interface	`CGJteamLab/Hilbert3DInterface.lean`
Coordinates/vectors	no
Euclidean parallel axiom (Group IV)	no
Continuity	no new use
New XI.4-specific axiom	no
Full project build	clean
Axiom audit	completed

4.20 Conclusion

Euclid XI.4 is the first major test of the project's spatial Hilbert architecture. The proposition begins with only two line-line perpendicularities at a common point and must end with a universal statement about every line of a plane through that point.

The formal proof shows that this transition can be reconstructed synthetically. The essential tools are not coordinates or vector orthogonality, but the same ingredients visible in Euclid's proof: equal-segment constructions, vertical angles, triangle congruence, an arbitrary transversal, and the definition of a perpendicular to a plane.

Lean exposes two pieces of structure that the classical diagram conceals. First, the arbitrary transversal requires a genuine Pasch argument with two connector families. Second, the repeated uses of Book I congruence in solid geometry require an explicit mechanism for comparing triangles carried by different planes.

Once these points are made explicit, the final statement is exactly Euclid XI.Def.3: the candidate line is perpendicular to every line of the given plane through the common point. In that sense the proof is not a modern replacement of XI.4 but a detailed synthetic reconstruction of its logical content.

Chapter 5

Proposition XI.5

5.1 Introduction

Proposition XI.5 is the first Book XI proposition whose conclusion is a genuinely spatial coplanarity statement forced by perpendicularity.

A line l passes through a point O and is perpendicular there to three lines m, n, p. Two of these lines, m and n, are assumed distinct. The conclusion is that m, n, p lie in one plane.

The formal reconstruction is not a line-by-line transcription of Euclid's proof. The current spatial interface makes it possible to use Proposition XI.4 twice and thereby obtain a short contradiction argument.

5.2 Euclid's Statement

Euclid's Proposition XI.5 states:

> If a straight line is set up at right angles to three straight lines which meet one another at their common point of section, then the three straight lines lie in one plane.

In Euclid's notation the common perpendicular is AB, and the three lines through the common point B are BC, BD, and BE.

5.3 Classical Configuration

Euclid assumes, for contradiction, that two of the three lines, say BD and BE, lie in a reference plane while BC lies in a more elevated plane.

He then considers the plane through AB and BC. Its intersection with the reference plane is a line BF. Thus AB, BC, BF lie in one plane.

At the same time, since AB is perpendicular to BD and BE, Proposition XI.4 implies that AB is perpendicular to the reference plane. Therefore AB is also perpendicular to BF.

The contradiction is that, in the plane through AB and BC, the lines BC and BF would both form a right angle with AB at the same point.

5.4 Classical Proof

The local classical dependency pattern is:

$$\text{XI.3} \longrightarrow \text{intersection line } BF,$$

$$\text{XI.4} \longrightarrow AB \perp \text{(reference plane)},$$

followed by Euclid's Definition XI.3, which turns line–plane perpendicularity into perpendicularity to every line of the plane meeting the perpendicular.

Thus the classical proof is a contradiction proof based on the intersection of two planes and on the impossibility of two distinct perpendiculars to the same line at the same point in one plane.

5.5 What is genuinely three-dimensional here?

The essential spatial content is not the elementary fact about right angles. It is the management of planes.

The proof must establish:

- a plane through two intersecting lines;
- a second plane through another pair of intersecting lines;
- that these two planes are distinct;
- a line of intersection of the two planes;
- containment of relevant lines in the appropriate planes.

The current Lean proof does not open a new planar `PlaneGeo` argument of its own. Instead, its planar geometric content is inherited through the already proved Proposition XI.4.

5.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_5
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
```

```
    [_HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (l m n p : Geo.Line)
    (O : Geo.Point)
    (hmn : Ne m n)
    (hPerpM :
      HilbertLinesPerpendicularAt Geo l m O)
    (hPerpN :
      HilbertLinesPerpendicularAt Geo l n O)
    (hPerpP :
      HilbertLinesPerpendicularAt Geo l p O) :
    exists pi : S.Plane,
      HilbertLineInPlane Geo m pi /\
      HilbertLineInPlane Geo n pi /\
      HilbertLineInPlane Geo p pi
```

The conclusion is constructive at the level of incidence: Lean returns an explicit plane π containing all three lines.

The hypothesis `hmn : Ne m n` makes explicit a point that is usually left implicit in the classical diagram. The two intersecting lines m, n must be distinct in order to determine the reference plane.

5.7 Spatial / Planar Architecture Used

The proposition uses the ambient three-dimensional layer:

- `HilbertSpacePrimitive` for planes and point–plane incidence;
- `HilbertSpaceIncidence` for spatial plane constructions;
- `HilbertLineInPlane` for line–plane containment;
- `HilbertLinesPerpendicularAt` for spatial line–line perpendicularity;
- `HilbertLinePerpendicularPlaneAt` for line–plane perpendicularity.

The ambient geometry does not install the planar order and congruence classes as global instances on the spatial object. The architectural distinction is:

```
-- not global ambient instances:
HilbertOrder Geo
HilbertCongruence Geo

-- spatial layers:
HilbertSpaceOrder
HilbertSpaceCongruence
```

Planar order and congruence enter only through the established `PlaneGeo` machinery used inside Proposition XI.4.

5.8 Proof Architecture of the Reconstruction

Let m and n be the first two lines perpendicular to l at O.

5.8.1 Step 1: construct the reference plane

Since $m \neq n$ and both pass through O, the theorem

```
hilbert_plane_through_two_intersecting_lines
```

constructs a plane π containing m and n.

5.8.2 Step 2: apply Proposition XI.4

The lines m and n are packaged as `PlaneLine`s of π, and O as a `PlanePoint` of π.

Proposition XI.4 then yields

$$l \perp \pi \quad \text{at } O.$$

In Lean this is the call to

```
euclid_proposition_11_4
```

with the two given perpendicularities $l \perp m$ and $l \perp n$.

5.8.3 Step 3: split on whether p already lies in π

If

$$p \subset \pi,$$

the theorem is finished immediately.

Assume therefore

$$p \not\subset \pi.$$

Since $l \perp p$, the lines l and p are distinct. They meet at O, so they determine another plane σ.

5.8.4 Step 4: intersect the two planes

The planes π and σ must be distinct. If $\pi = \sigma$, then $p \subset \sigma$ would imply $p \subset \pi$, contrary to the case assumption.

The theorem

```
hilbert_plane_intersection_line
```

therefore gives a line q through O contained in both planes:

$$q \subset \pi, \qquad q \subset \sigma.$$

Since $l \perp \pi$ and q is a line of π through O,

$$l \perp q.$$

This is obtained directly from

```
HilbertLinePerpendicularPlaneAt.perpendicular_to_line
```

5.8.5 Step 5: the second use of XI.4

Now p and q both lie in σ, and

$$l \perp p, \qquad l \perp q.$$

Suppose $p \neq q$. Applying Proposition XI.4 inside σ gives

$$l \perp \sigma.$$

But $l \subset \sigma$ by construction. This is impossible.

Hence

$$p = q.$$

Since $q \subset \pi$, it follows that $p \subset \pi$, and therefore

$$m, n, p \subset \pi.$$

5.9 Lean Formalization

The small proposition-local helper is:

```
theorem hilbert_linePerpendicularPlaneAt_not_in_plane
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    (l : Geo.Line)
    (pi : S.Plane)
    (O : Geo.Point)
```

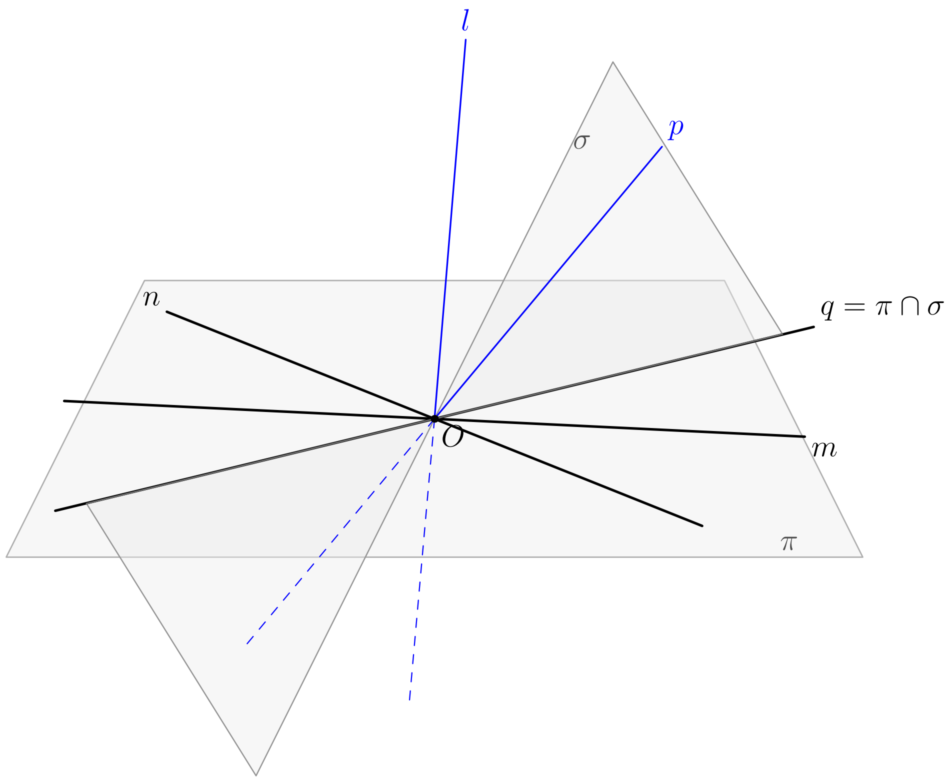

Figure 5.1: The formal contradiction argument for Proposition XI.5. The black lines m, n, q lie in the reference plane π. The spatial lines l, p are blue. Under the temporary assumption $p \not\subset \pi$, the lines l, p determine the second plane σ, and $q = \pi \cap \sigma$. If $p \neq q$, a second application of XI.4 would make l perpendicular to σ, although $l \subset \sigma$.

```
    (hPerp :
      HilbertLinePerpendicularPlaneAt Geo l pi O) :
    Not (HilbertLineInPlane Geo l pi) := by

  intro hlpi

  have hSelf :
      HilbertLinesPerpendicularAt Geo l l O :=
    HilbertLinePerpendicularPlaneAt.perpendicular_to_line
      (Geo := Geo)
      hPerp
      hlpi
      hPerp.1

  exact
    (hilbert_linesPerpendicularAt_ne
      Geo l l O hSelf) rfl
```

This helper is purely definitional in geometric content. If a line perpendicular to a plane were itself contained in the plane, then the universal line–plane perpendicularity condition would make the line perpendicular to itself. The predicate `HilbertLinesPerpendicularAt` is nondegenerate, so this is impossible.

The theorem body then consists of two plane constructions, one plane-intersection construction, and two calls to XI.4.

5.10 Hidden Formal Data

Several facts visible in a classical spatial diagram must be supplied explicitly in Lean.

- O lies in the plane determined by m, n.
- The ambient lines m, n must be packaged as `PlaneLine Geo pi`.
- Their distinctness must be transported to the subtype level.
- If $p \not\subset \pi$, the plane through l, p must be constructed explicitly.
- The two planes must be proved distinct before their intersection line can be used.
- The intersection line q must carry proofs that it lies in both π and σ.
- For the second XI.4 call, p, q and O are repackaged as `PlaneLine` and `PlanePoint` objects of σ.

These are mathematical incidence obligations, not merely syntactic casts. They encode information that Euclid's diagram leaves implicit.

5.11 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses XI.3 and XI.4.

The Lean proof imports Proposition XI.4 and uses the current spatial intersection theorem

```
hilbert_plane_intersection_line
```

instead of importing a separate formal Proposition XI.3.

Thus the two dependency graphs are not identical.

The actual Lean Book XI dependency is:

$$\boxed{\text{XI.4}} \longrightarrow \boxed{\text{XI.5}}.$$

Proposition XI.5 is then the coplanarity tool required by the classical route to Proposition XI.6.

5.12 Relationship with Book I / Book Zero / Hilbert Infrastructure

No Book I proposition is called directly by the XI.5 theorem body.

The inherited planar Hilbert and Book Zero infrastructure enters transitively through Proposition XI.4. In particular, XI.4 internally uses induced plane geometry, while XI.5 itself is primarily a spatial incidence argument.

The new helper `hilbert_linePerpendicularPlaneAt_not_in_plane` does not add a geometric axiom; it follows from the existing definitions and nondegeneracy of line–line perpendicularity.

5.13 Axiomatic Status

The audit command is:

```
import CGJteamLab.Proposition11_5

#print axioms Geometry.euclid_proposition_11_5
```

The reported dependency set is:

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

Therefore:

- new proposition-specific geometric axiom: no;
- new spatial axiom: no;
- new continuity assumption: no;
- Euclidean parallel axiom / Hilbert Group IV: not used;

- new proposition-local helper: yes;
- new `Hilbert3DAxioms` declaration: no;
- new `PlaneGeo` bridge theorem: no;
- `sorry`/`admit`: no.

The inherited dependency `Geometry.bookZero_nullSegment1` is the same accepted dependency already present in Proposition XI.4.

5.14 Dependency Summary

The actual formal dependency chain is:

Hilbert incidence / order / congruence infrastructure
↓
spatial primitive / incidence / order / congruence
↓
PlaneGeo and plane–space bridges used by XI.4
↓
`euclid_proposition_11_4`
↓
plane through intersecting lines
intersection line of two planes
`hilbert_linePerpendicularPlaneAt_not_in_plane`
↓
`euclid_proposition_11_5`.

At the current checkpoint the production theorem compiles locally and its axiom audit has been run successfully. A full repository `lake build` is a separate checkpoint and should be recorded only after it is run.

5.15 Conclusion

Proposition XI.5 identifies the plane perpendicular to a given line at a given point as the unique carrier of all lines through that point perpendicular to the line, in the precise three-line form stated by Euclid.

The formal proof is shorter than the classical local proof graph because the present 3D interface already contains an intersection-line theorem for two planes and because XI.4 can be applied a second time to exclude the hypothetical second plane.

No coordinates, vectors, scalar products, continuity principle, or new spatial axiom enter the proof.

Chapter 6

Proposition XI.6

6.1 Introduction

Proposition XI.6 is the first proposition in the present Book XI development whose conclusion is a genuinely spatial parallelism statement. Two ambient lines l and m are perpendicular to the same plane π, at distinct feet B and D. Euclid concludes that the two lines are parallel.

In three-dimensional geometry this conclusion contains two logically different assertions. The lines must first be shown to be *coplanar*; only then can planar geometry prove that they do not meet. This distinction is essential in the formal reconstruction, because two disjoint ambient lines may be skew.

The current Lean proof follows the metric core of Euclid's construction closely:

$$\text{auxiliary perpendicular and equal segment} \longrightarrow \text{SAS} \longrightarrow \text{SSS} \longrightarrow \text{XI.5}.$$

After XI.5 has supplied the missing common plane, the proof changes deliberately to an induced planar geometry and closes the disjointness argument by the neutral alternate-angle criterion.

6.2 Euclid's Statement

Euclid's Proposition XI.6 states:

> If two straight lines be at right angles to the same plane, the straight lines will be parallel.

In the classical configuration the two perpendiculars are AB and CD, meeting the reference plane at B and D. The formal theorem makes the nondegenerate case explicit by assuming

$$B \neq D.$$

6.3 Classical Configuration

Join the feet B and D. In the reference plane construct through D a line DE perpendicular to BD, and choose E so that

$$DE \cong AB.$$

Join BE, AE, and AD.

Because AB is perpendicular to the reference plane, it is perpendicular to every line of that plane through B, in particular to BD and BE. Hence

$$\angle ABD, \qquad \angle ABE$$

are right angles.

Likewise CD is perpendicular to the reference plane at D, so it is perpendicular to both DB and DE.

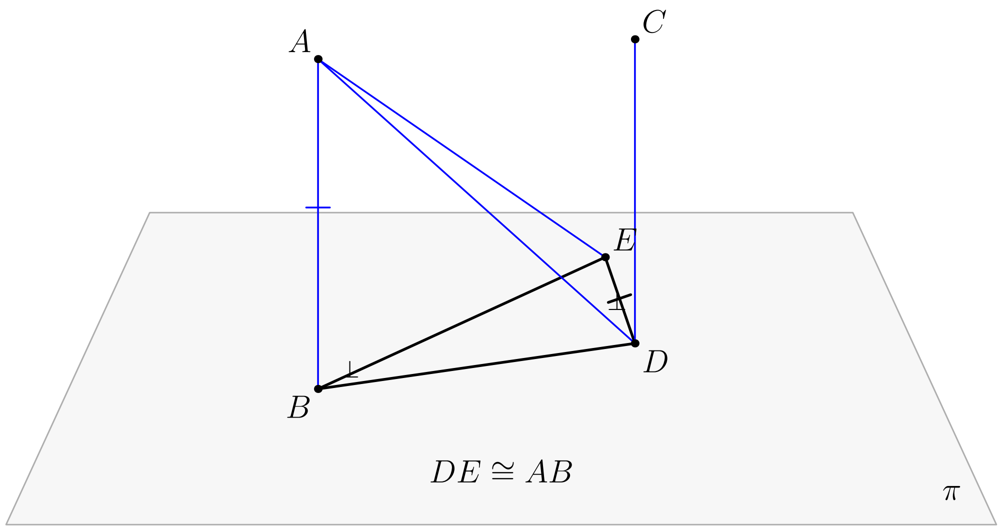

Figure 6.1: The classical configuration of Proposition XI.6. The black geometry lies in the reference plane π: B, D, E, the transversal BD, and the auxiliary perpendicular DE. The blue lines and connectors leave π. The construction $DE \cong AB$ prepares the SAS comparison of BAD and DEB; the resulting equality $AD \cong EB$ prepares the SSS comparison of DEA and BAE.

6.4 Classical Proof

The classical metric part consists of two triangle-congruence steps.

First,

$$AB \cong DE, \qquad BD \cong DB,$$

and the included angles

$$\angle ABD, \qquad \angle EDB$$

are both right. Proposition I.4 therefore gives

$$AD \cong EB.$$

Second, the triangles DEA and BAE have

$$DE \cong BA, \qquad EA \cong AE, \qquad DA \cong BE.$$

Proposition I.8 gives

$$\angle EDA \cong \angle ABE.$$

Since $\angle ABE$ is right, $\angle EDA$ is right; hence

$$ED \perp DA.$$

Now at the point D the line ED is perpendicular to the three lines

$$DB, \qquad DA, \qquad DC.$$

Proposition XI.5 therefore places those three lines in one plane. Proposition XI.2 identifies the triangle plane determined by the intersecting lines, so AB is brought into the same plane as BD and DC. Finally

$$\angle ABD = \angle BDC = 90^\circ,$$

and Proposition I.28 yields

$$AB \parallel CD.$$

A compact classical dependency graph is therefore

$$\begin{gathered}
\text{construct } DE \perp BD, \quad DE \cong AB \\
\Downarrow \\
\text{I.4} \\
\Downarrow \\
AD \cong EB \\
\Downarrow \\
\text{I.8} \\
\Downarrow \\
ED \perp DA \\
\Downarrow \\
\text{XI.5} \\
\Downarrow \\
DB, DA, DC \text{ coplanar} \\
\Downarrow \\
\text{XI.2} \\
\Downarrow \\
AB, BD, DC \text{ coplanar} \\
\Downarrow \\
\text{I.28} \\
\Downarrow \\
AB \parallel CD.
\end{gathered}$$

The construction of the auxiliary perpendicular and equal segment is part of the earlier Book I construction machinery. The substantive local congruence steps are I.4 and I.8.

6.5 What is genuinely three-dimensional here?

The difficult point is not the planar theorem that two perpendiculars to one transversal are parallel. That statement cannot even be applied until one knows that the two candidate parallel lines lie in one plane.

In ambient three-space the information

$$l \perp BD, \qquad m \perp BD$$

does not by itself constitute a parallelism proof: without a coplanarity theorem the lines could still be skew.

The formal proof therefore separates XI.6 into

$$\boxed{\text{coplanarity}} \quad + \quad \boxed{\text{planar disjointness}}.$$

The first box is the genuinely spatial content and is exactly where Proposition XI.5 is used.

6.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_6
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    [_HSO : HilbertSpaceOrder
      (Geo := Geo) (H := H) (S := S)]
    [HSC : HilbertSpaceCongruence
      (Geo := Geo) (H := H) (S := S)]
    (pi : S.Plane)
    (l m : Geo.Line)
    (B D : Geo.Point)
    (hBD : Ne B D)
    (hLperp :
      HilbertLinePerpendicularPlaneAt
        Geo l pi B)
    (hMperp :
      HilbertLinePerpendicularPlaneAt
        Geo m pi D) :
    HilbertSpaceLinesParallel Geo l m
```

The target relation is part of the reusable spatial interface `Hilbert3DInterface`:

```
def HilbertSpaceLinesParallel
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    (l m : Geo.Line) : Prop :=
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
    HilbertLinesDisjoint Geo l m
```

Thus "parallel in space" means exactly:

1. there is a plane containing both lines;
2. the two lines are disjoint.

The first clause is not redundant. If spatial parallelism were defined only by ambient disjointness, skew lines would incorrectly satisfy the definition.

6.7 Spatial / Planar Architecture Used

The proof uses three layers which must remain distinct.

6.7.1 Ambient three-space

The ambient objects and relations include

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
HilbertLineInPlane
HilbertLinesPerpendicularAt
HilbertLinePerpendicularPlaneAt
```

In particular, segment congruence between configurations carried by different planes is handled through `HilbertSpaceCongruence`.

6.7.2 Induced plane geometry

Whenever a planar theorem is needed, the relevant ambient plane σ or π is converted into

```
PlanePoint
PlaneLine
PlaneGeo
```

and ordinary planar incidence, order, right-angle, side-of-line, and triangle theorems are used there.

There is deliberately no global ambient instance

```
HilbertOrder Geo
HilbertCongruence Geo
```

for the three-dimensional space.

6.7.3 Plane–space bridges

The proof repeatedly uses bridges such as

```
planeGeo_linesPerpendicularAt_iff_ambient
planeGeo_rightAngle_iff_ambient
planeGeo_angleCongruent_iff_ambient
planeGeo_primCollinear_to_ambient
planeGeo_not_primCollinear_to_ambient
planeGeo_sameRay_iff_ambient
```

These are not cosmetic casts. They state which geometric fact is being used in which plane slice.

6.8 Proof Architecture of the Reconstruction

6.8.1 Step 1: Euclid's auxiliary construction in the reference plane

Choose a point $A \neq B$ on the first normal l. Inside the reference plane π, construct the line

$$d = BD.$$

Through D construct a line e perpendicular to d, and lay off on e a point E such that

$$DE \cong AB.$$

This is packaged by

```
hilbert_XI6_auxiliary_perpendicular_equal_segment
```

The line d and the auxiliary line e are constructed in `PlaneGeo pi`. The segment copied onto DE, however, may have the spatial endpoint A; that copy is therefore performed by `HilbertSpaceCongruence.segment_construction`.

6.8.2 Step 2: the first spatial SAS

Since $l \perp \pi$ at B and $d \subset \pi$ passes through B,

$$l \perp d.$$

The auxiliary construction gives

$$e \perp d$$

at D.

The helper

```
hilbert_XI6_space_linesPerpendicularAt_right_angle_of_points
```

normalizes these line–line perpendicularities to the concrete right angles

$$\angle ABD, \qquad \angle EDB.$$

The spatial right-angle comparison then supplies

$$\angle ABD \cong \angle EDB.$$

With

$$BA \cong DE, \qquad BD \cong DB,$$

the spatial SAS theorem gives

$$AD \cong EB.$$

This whole block is packaged by

```
hilbert_XI6_first_SAS
```

and uses

```
hilbert_space_sas_third_side_and_angle
```

for the actual triangle comparison.

6.8.3 Step 3: the second spatial SSS and the new right angle

The next comparison is between the triangles DEA and BAE. The three side relations are

$$DE \cong BA, \qquad EA \cong AE, \qquad DA \cong BE.$$

The spatial SSS theorem gives

$$\angle EDA \cong \angle ABE.$$

Because $AB \perp \pi$ and $BE \subset \pi$, the angle ABE is right. The rightness is transported to EDA, yielding

$$ED \perp DA.$$

This block is packaged by

```
hilbert_XI6_second_SSS_right_angle
```

The proof must create explicit carrier planes for the relevant triangles. The right-angle transport is handled by

```
hilbert_XI6_space_right_angle_transport_in_plane
```

rather than by installing a forbidden ambient `HilbertCongruence Geo` instance.

6.8.4 Step 4: Proposition XI.5 supplies coplanarity

At D, the line $e = DE$ is now perpendicular to

$$d = DB, \qquad a = DA, \qquad m.$$

The last perpendicularity follows because $m \perp \pi$ and $e \subset \pi$.

Proposition XI.5 is applied exactly here:

```
euclid_proposition_11_5
```

It produces a plane σ containing

$$d, \qquad a, \qquad m.$$

Since $A \in a$ and $B \in d$, both A and B lie in σ. The line $l = AB$ therefore also lies in σ.

The result is isolated as

```
hilbert_XI6_normals_to_same_plane_coplanar
```

with conclusion

```
exists sigma : S.Plane,
  HilbertLineInPlane Geo l sigma /\
  HilbertLineInPlane Geo m sigma
```

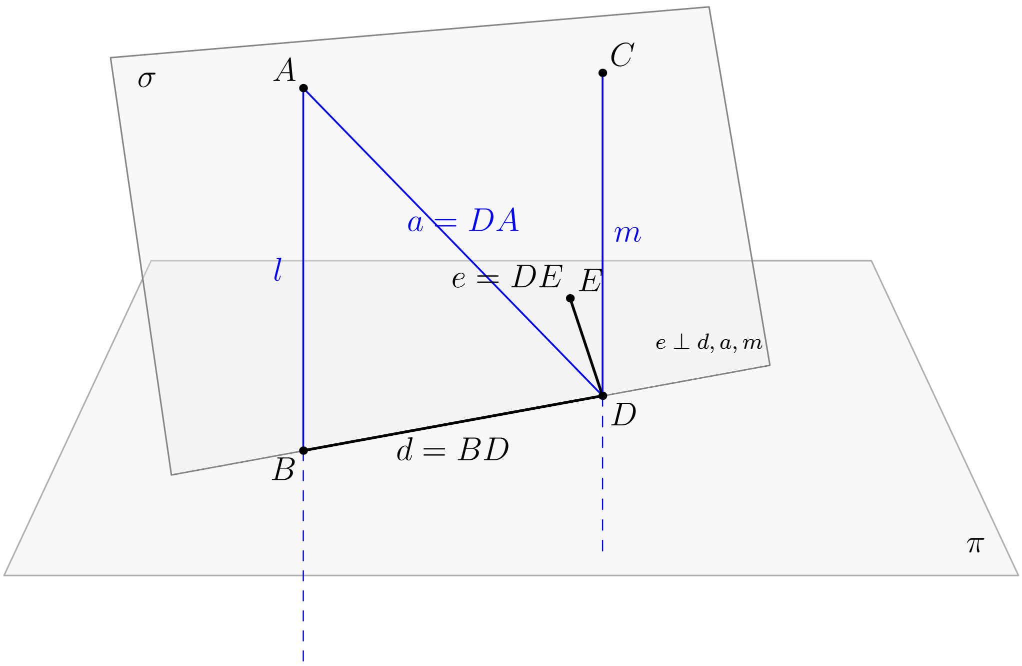

Figure 6.2: The genuinely spatial step in the formal XI.6 proof. The black lines $d = BD$ and $e = DE$ lie in the reference plane π. The blue lines $a = DA$, $m = DC$, and $l = AB$ leave π. At D, the line e is perpendicular to d, a, m, so XI.5 produces the second plane σ containing d, a, m. Because σ contains A and B, it also contains l. The lines l and m are therefore coplanar.

6.8.5 Step 5: the remaining problem is planar

After Step 4, both l and m lie in σ. The common transversal $d = BD$ lies in σ as well, and

$$l \perp d \text{ at } B, \qquad m \perp d \text{ at } D.$$

The rest of the proof is carried out entirely inside

$$\texttt{PlaneGeo Geo sigma}.$$

The helper

```
hilbert_XI6_coplanar_normals_disjoint
```

chooses nonfoot points on l and m, establishes the required side-of-line orientation relative to d, compares the two right angles, and applies

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

to obtain planar parallelism.

The final conversion proves that the actual ambient carriers l and m are disjoint.

6.8.6 Step 6: assemble spatial parallelism

The public theorem now has exactly the two pieces required by `HilbertSpaceLinesParallel`:

$$l, m \subset \sigma$$

and

$$\text{HilbertLinesDisjoint}(l, m).$$

Therefore

$$\text{HilbertSpaceLinesParallel}(l, m).$$

6.9 Lean Formalization

The final theorem body is short because the geometric work has been isolated into two major helpers. First,

```
hilbert_XI6_normals_to_same_plane_coplanar
```

returns a plane `sigma` containing both ambient normals. Then

```
hilbert_XI6_coplanar_normals_disjoint
```

proves that their ambient carriers are disjoint. The public theorem packages exactly these three fields of `HilbertSpaceLinesParallel`: containment of l, containment of m, and disjointness.

The production module imports

```
import CGJteamLab.Proposition11_5
import CGJteamLab.Proposition27
```

The first import supplies the decisive spatial coplanarity theorem. The second supplies the neutral alternate-angle machinery used in the final induced plane.

6.10 Hidden Formal Data

The classical diagram suppresses several obligations which the Lean proof must make explicit.

- The feet B, D are distinct.
- The point A selected on l is outside the reference plane π.
- The auxiliary point E is nondegenerate and lies both on e and in π.
- Every triangle used in SAS or SSS carries an explicit noncollinearity proof.
- Spatial triangle comparisons require an explicit target carrier plane.
- Right-angle transport between ambient configurations is performed through an explicit `PlaneGeo`, not through an ambient planar congruence instance.
- Before XI.5 can be applied, the three lines d, a, m must be proved distinct in the required places and must all pass through D.
- The plane σ returned by XI.5 must be shown to contain both A and B, so that the whole carrier $l = AB$ lies in σ.
- In the final planar proof, the chosen points on l and m must be oriented on opposite sides of the transversal d. The proof performs an explicit SameSide/OppositeSide case split.
- Point-pair parallelism inside `PlaneGeo sigma` must finally be converted into disjointness of the original ambient carriers.

These obligations are geometric. They are the incidence, order, and plane-membership data that a conventional three-dimensional drawing usually leaves implicit.

6.11 Relationship with Earlier Book XI Propositions

Proposition XI.5 is a direct formal dependency:

$$\boxed{\text{XI.5}} \longrightarrow \boxed{\text{XI.6}}.$$

XI.5 itself depends on XI.4, and XI.6 also reuses several XI.4 normalization helpers for perpendicular lines and right angles. Consequently the effective Book XI chain is

$$\boxed{\text{XI.4}} \longrightarrow \boxed{\text{XI.5}} \longrightarrow \boxed{\text{XI.6}}.$$

The classical proof also invokes XI.2 when it moves from the three coplanar lines through D to the plane containing the triangle ABD. The Lean proof does not need a separate formal call to Proposition XI.2: the spatial incidence API directly constructs and tracks the relevant carrier plane.

6.12 Relationship with Book I / Book Zero / Hilbert Infrastructure

The metric part of the formal proof reproduces the roles of Euclid I.4 and I.8 through the spatial SAS and SSS infrastructure:

```
hilbert_space_sas_third_side_and_angle
hilbert_space_sss_angleA_in_plane
```

Right angles and their transport are handled synthetically; no angle measure is introduced.

The classical final step is I.28. The production Lean file instead imports `Proposition27` and uses the established neutral alternate-angle theorem

```
hilbert_parallel_of_alternate_angles_oppositeSide_lines
```

inside the explicitly constructed plane σ. Thus the formal module dependency and the classical Euclid proposition-number dependency are not identical, even though they close the same planar configuration.

Book Zero enters only transitively through the established Hilbert geometric infrastructure. No new Book Zero axiom or theorem is added by XI.6.

6.13 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_6

#print axioms Geometry.euclid_proposition_11_6
```

and Lean reports

```
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

This is the same reported axiom profile as Propositions XI.4 and XI.5.

Therefore:

- new proposition-specific geometric axiom: no;
- new spatial axiom: no;
- new continuity assumption: no;
- Euclidean parallel axiom / Hilbert Group IV: not used;
- new proposition-local helper theorems: yes;
- new `Hilbert3DAxioms` declaration: no;
- reusable `Hilbert3DInterface` declaration: `HilbertSpaceLinesParallel`, promoted from the XI.6 proposition module for reuse by XI.7 and later spatial results;

- new `PlaneGeo` bridge theorem: no;
- `sorry`/`admit`: no.

The inherited `Geometry.bookZero_nullSegment1` dependency is not specific to XI.6.

The production theorem compiled successfully, its axiom audit was run successfully, and the full repository `lake build` was reported clean at this checkpoint.

6.14 Dependency Summary

The classical local chain is

$$\begin{array}{c}\text{Book I perpendicular/equal-segment construction}\\ \downarrow \\ \text{I.4 (SAS)}\\ \downarrow \\ \text{I.8 (SSS)}\\ \downarrow \\ \text{XI.5}\\ \downarrow \\ \text{XI.2}\\ \downarrow \\ \text{I.28}\\ \downarrow \\ \text{XI.6.}\end{array}$$

The actual formal chain is

$$
\begin{gathered}
\text{Hilbert incidence / order / congruence infrastructure}\\
\downarrow\\
\text{spatial primitive / incidence / order / congruence}\\
\downarrow\\
\text{PlaneGeo slices and plane–space bridges}\\
\downarrow\\
\text{XI.4 right-angle and perpendicular normalizers}\\
\downarrow\\
\text{auxiliary } d = BD,\ e \perp d,\ DE \cong AB\\
\downarrow\\
\text{spatial SAS: } AD \cong EB\\
\downarrow\\
\text{spatial SSS: } ED \perp DA\\
\downarrow\\
\texttt{euclid\_proposition\_11\_5}\\
\downarrow\\
l, m \text{ coplanar in } \sigma\\
\downarrow\\
\text{PlaneGeo}(\sigma) \text{ side/order + alternate angles}\\
\downarrow\\
\text{ambient carrier disjointness}\\
\downarrow\\
\texttt{HilbertSpaceLinesParallel}\\
\downarrow\\
\texttt{euclid\_proposition\_11\_6}.
\end{gathered}
$$

6.15 Conclusion

Proposition XI.6 is a useful boundary theorem for the three-dimensional architecture. Its metric core is recognizably Euclidean: construct an auxiliary perpendicular and an equal segment, use SAS, then SSS, and obtain a third perpendicular at D.

The genuinely spatial step is XI.5. It converts three perpendicularities through one point into the common plane in which the final parallelism argument is allowed to take place. Lean makes this logical boundary explicit rather than allowing the diagram to hide it.

The final definition of spatial parallelism records the same distinction: coplanarity is proved first and disjointness second. No coordinates, vectors, scalar products, numerical angle measure, continuity principle, new spatial axiom, or Euclidean parallel axiom is introduced.

Chapter 7

Proposition XI.7

7.1 Introduction

Proposition XI.7 is an incidence proposition about spatial parallelism. Two ambient straight lines are parallel, points are chosen on the two lines, and Euclid asserts that the straight line joining the chosen points lies in the same plane as the parallels.

The formal reconstruction is substantially shorter than XI.4–XI.6. The reason is structural. In the current spatial interface, `HilbertSpaceLinesParallel` already contains an explicit common carrier plane. Once the two chosen points are shown to lie in that plane, Hilbert's spatial incidence axiom I.6 places their joining line in the same plane.

Thus the formal proof is not a line-by-line translation of Euclid's reductio. It proves the same geometric target through the stronger and more explicit incidence organization adopted by Hilbert.

7.2 Euclid's Statement

Euclid's Proposition XI.7 states:

> If two straight lines be parallel and points be taken at random on each of them, the straight line joining the points is in the same plane with the parallel straight lines.

Let AB and CD be parallel straight lines. Choose E on AB and F on CD. The claim is that the joining straight line EF lies in the plane of AB and CD.

7.3 Classical Configuration

The classical configuration begins with the plane containing the two parallel lines AB and CD. The points E and F are arbitrary points on the two lines.

The proposition is genuinely spatial not because the final figure contains a line outside the plane, but because the assertion excludes that possibility.

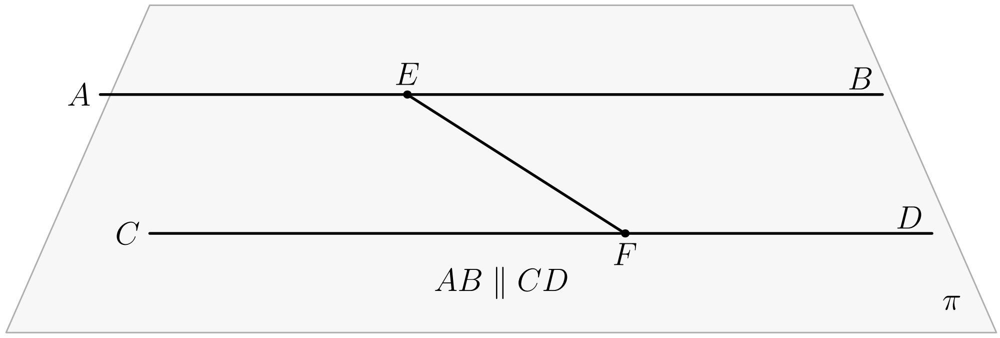

Figure 7.1: The configuration of Proposition XI.7. The two parallel lines AB and CD lie in the reference plane π, with $E \in AB$ and $F \in CD$. The proposition asserts that the joining line EF also lies in π. All final geometric objects in the figure are therefore drawn in black; no object in the proved configuration leaves the plane.

7.4 Classical Proof

Euclid argues by contradiction. Suppose that the straight line joining E and F does not lie in the plane of the parallels, but in a more elevated plane. Draw a plane through that alleged elevated straight line.

The new plane cuts the plane of the parallel lines in a straight line, by Proposition XI.3. Since both E and F belong to both planes, the common section passes through E and F. Euclid then obtains two distinct straight lines through the same two points, described in the text as two straight lines enclosing an area, which is impossible.

Hence the joining line EF cannot leave the plane of the parallels.

The local classical DAG is therefore

$$
\begin{gathered}
AB \parallel CD, \quad E \in AB, \quad F \in CD \\
\Downarrow \\
\text{suppose the join } EF \text{ leaves the plane of the parallels} \\
\Downarrow \\
\text{draw a second plane through the alleged elevated join} \\
\Downarrow \\
\text{XI.3} \\
\Downarrow \\
\text{the two planes have a straight-line section through } E, F \\
\Downarrow \\
\text{two distinct straight lines through } E, F \\
\Downarrow \\
\text{impossible} \\
\Downarrow \\
EF \text{ lies in the plane of } AB, CD.
\end{gathered}
$$

The only explicit Book XI proposition cited in the proof is XI.3.

7.5 What is genuinely three-dimensional here?

The genuinely spatial datum is the carrier plane of the two parallel ambient lines. In ordinary plane geometry this datum is invisible, because every line under discussion already belongs to the unique ambient plane.

In three-space, disjointness alone is insufficient for parallelism: two disjoint lines may be skew. The formal relation `HilbertSpaceLinesParallel` therefore records both

$$\text{coplanarity} \qquad \text{and} \qquad \text{disjointness.}$$

XI.7 uses the coplanarity component directly. No metric geometry is needed.

7.6 Formal Statement

The production theorem is:

```
theorem euclid_proposition_11_7
    [H : HilbertIncidence Geo]
    [HilbertPlaneIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    [HSI : HilbertSpaceIncidence Geo]
    (l m : Geo.Line)
    (E F : Geo.Point)
    (hParallel : HilbertSpaceLinesParallel Geo l m)
    (hEl : H.OnLine E l)
    (hFm : H.OnLine F m) :
    exists sigma : S.Plane,
      HilbertLineInPlane Geo l sigma /\
      HilbertLineInPlane Geo m sigma /\
      exists n : Geo.Line,
        H.OnLine E n /\
        H.OnLine F n /\
        HilbertLineInPlane Geo n sigma
```

The conclusion returns both the common carrier plane `sigma` and an explicit joining line `n` through E and F, together with the proof that `n` lies in `sigma`.

7.7 Spatial / Planar Architecture Used

XI.7 uses only the ambient incidence layer:

```
HilbertIncidence
HilbertPlaneIncidence
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertLineInPlane
HilbertLinesDisjoint
HilbertSpaceLinesParallel
```

No induced `PlaneGeo` is constructed. There is no planar betweenness, SameRay, SameSide/OppositeSide, angle congruence, right angle, SAS, SSS, or plane–space congruence transport.

This is an important contrast with XI.4–XI.6: although XI.7 is a spatial proposition, its formal content is pure incidence.

7.8 Proof Architecture of the Reconstruction

7.8.1 Step 1: unpack spatial parallelism

From

$$\text{HilbertSpaceLinesParallel}(l, m)$$

the proof obtains a plane σ such that

$$l \subset \sigma, \qquad m \subset \sigma,$$

together with

$$\text{HilbertLinesDisjoint}(l, m).$$

The definition is now part of the general `Hilbert3DInterface` rather than a proposition-local declaration.

7.8.2 Step 2: place the chosen points in the common plane

Since $E \in l$ and $l \subset \sigma$,

$$E \in \sigma.$$

Likewise $F \in m$ and $m \subset \sigma$, hence

$$F \in \sigma.$$

These are direct applications of the line-in-plane predicates supplied by the parallelism witness.

7.8.3 Step 3: prove that the chosen points are distinct

The joining line construction requires $E \neq F$. Euclid's diagram does not state this separately, but the formal proof derives it from the disjointness of the two parallel carriers.

If $E = F$, then the common point would lie on both l and m, contradicting

$$\text{HilbertLinesDisjoint}(l, m).$$

Thus

$$E \neq F.$$

7.8.4 Step 4: construct the joining line

The ordinary incidence structure supplies a line through the two distinct points:

```
HilbertPlaneIncidence.line_through E F hEF
```

This yields a line n with
$$E \in n, \qquad F \in n.$$

7.8.5 Step 5: apply Hilbert I.6

Now E and F are two distinct points of n, and both lie in σ. The decisive step is:

```
HilbertSpaceIncidence.line_in_plane
```

which gives
$$n \subset \sigma.$$

This is the direct formal counterpart of Hilbert's incidence axiom I.6: two points of a line lying in a plane force the whole line into that plane.

7.9 Lean Formalization

The proof DAG is extremely short:

$$\begin{gathered}
\texttt{HilbertSpaceLinesParallel}\ l\ m \\
\Downarrow \\
\exists \sigma, \quad l \subset \sigma, \quad m \subset \sigma, \quad l \cap m = \varnothing \\
\Downarrow \\
E \in \sigma, \quad F \in \sigma \\
\Downarrow \\
E \neq F \\
\Downarrow \\
\texttt{line_through}\ E\ F \\
\Downarrow \\
n \text{ through } E, F \\
\Downarrow \\
\texttt{HilbertSpaceIncidence.line_in_plane} \\
\Downarrow \\
n \subset \sigma \\
\Downarrow \\
\texttt{euclid_proposition_11_7}.
\end{gathered}$$

There are no proposition-local helper theorems.

7.10 Hidden Formal Data

XI.7 has very little hidden data, but two obligations must be explicit.

- The arbitrary points E and F are distinct. This follows from disjointness of the two parallel carriers.
- The phrase "the straight line joining E and F" is represented by an explicit existential line witness produced by `HilbertPlaneIncidence.line_through`.

Once those two points are available, Hilbert I.6 closes the spatial incidence problem directly.

7.11 Relationship with Earlier Book XI Propositions

The classical proof explicitly uses Proposition XI.3, because Euclid constructs a second plane and studies the straight line in which the two planes intersect.

The current Lean theorem does *not* import or invoke XI.3, XI.4, XI.5, or XI.6. Its production file imports only

```
import CGJteamLab.Hilbert3DInterface
```

This difference is structural rather than geometric. Euclid proves the plane containment by contradiction from a plane-intersection theorem. Hilbert's incidence system takes the closure of a line under two plane points as an explicit spatial incidence axiom.

XI.6 is nevertheless conceptually adjacent. It produces `HilbertSpaceLinesParallel`, which is exactly the hypothesis consumed by XI.7. The theorem XI.7 itself, however, is independent of the proof of XI.6.

7.12 Relationship with Book I / Hilbert Infrastructure

Euclid's definition of parallel straight lines includes coplanarity. The current formal definition makes this explicit:

```
def HilbertSpaceLinesParallel
    [H : HilbertIncidence Geo]
    [S : HilbertSpacePrimitive Geo]
    (l m : Geo.Line) : Prop :=
  exists sigma : S.Plane,
    HilbertLineInPlane Geo l sigma /\
    HilbertLineInPlane Geo m sigma /\
    HilbertLinesDisjoint Geo l m
```

For XI.7, the disjointness field is used only to derive $E \neq F$; the common-plane fields provide the main spatial information.

The decisive Hilbert source is Group I, Axiom I.6. In Hilbert's classification, I.1–I.3 are the plane incidence axioms and I.4–I.8 are the space incidence axioms. Thus the formal proof lands exactly in the spatial incidence layer and nowhere higher in the hierarchy.

7.13 Axiomatic Status

The audit command is

```
import CGJteamLab.Proposition11_7

#print axioms Geometry.euclid_proposition_11_7
```

and Lean reports:

```
'Geometry.euclid_proposition_11_7' does not depend on any axioms
```

This statement must be interpreted precisely. The theorem still has explicit typeclass hypotheses, including `HilbertSpaceIncidence Geo`. The audit says that its proof term does not depend on any additional global Lean axioms or project axiom constants beyond those explicit parameters.

Therefore:

- new proposition-specific geometric axiom: no;
- new spatial axiom: no;
- new continuity assumption: no;
- Euclidean parallel axiom / Hilbert Group IV: not used;
- spatial order: not used;
- spatial congruence: not used;
- induced `PlaneGeo`: not used;
- new proposition-local helper theorem: no;
- new `Hilbert3DAxioms` declaration: no;
- reusable `Hilbert3DInterface` change: yes, `HilbertSpaceLinesParallel` was promoted there;
- new `PlaneGeo` bridge theorem: no;
- `sorry`/`admit`: no.

The production theorem and the full repository build were reported clean at this checkpoint.

7.14 Dependency Summary

The classical local chain is

$$\begin{gathered} AB \parallel CD, \quad E \in AB, \quad F \in CD \\ \downarrow \\ \text{contradictory elevated joining line} \\ \downarrow \\ \text{XI.3} \\ \downarrow \\ \text{common section of two planes is a straight line} \\ \downarrow \\ \text{impossible double join through } E, F \\ \downarrow \\ \text{XI.7.} \end{gathered}$$

The actual formal chain is

$$
\begin{gathered}
\text{Hilbert incidence} \\
\downarrow \\
\text{Hilbert spatial incidence I.6} \\
\downarrow \\
\texttt{HilbertSpaceLinesParallel} \\
\downarrow \\
\text{common carrier plane } \sigma \\
\downarrow \\
E, F \in \sigma, \quad E \neq F \\
\downarrow \\
\texttt{HilbertPlaneIncidence.line\_through} \\
\downarrow \\
\texttt{HilbertSpaceIncidence.line\_in\_plane} \\
\downarrow \\
\texttt{euclid\_proposition\_11\_7}.
\end{gathered}
$$

7.15 Conclusion

Proposition XI.7 is a useful contrast with the preceding Book XI formalizations. Its statement is spatial, but its proof requires no order, congruence, right angles, induced plane geometry, or earlier Book XI theorem.

Euclid proves the result by a reductio involving a second plane and Proposition XI.3. The Hilbert reconstruction exposes the same geometric content at the incidence level: spatial parallelism already supplies the common plane, and Axiom I.6 says that a line with two distinct points in that plane lies wholly in the plane.

The resulting Lean theorem is correspondingly small. Its axiom audit reports no global axiomatic dependencies, while the required spatial incidence assumptions remain explicit theorem parameters. No coordinates, vectors, scalar products, numerical angle measures, continuity principles, or Euclidean parallel axiom are introduced.

Appendix A

Euclid in Heath's Edition: Source Audit

A.1 Scope

This appendix records the Book XI source audit against Euclid's *Elements* in Heath's edition through Proposition XI.7 [1]. The purpose is to control the classical statement, construction, and local proposition dependencies. Lean-specific incidence, subtype, and plane-transport obligations are documented in the proposition chapters and are not attributed to Euclid or Heath.

A.2 Proposition XI.4

Proposition XI.4 states that if a straight line is set up at right angles to each of two straight lines at their point of intersection, then it is also at right angles to the plane through those two lines.

The classical proof chooses equal segments on the two given lines and introduces an arbitrary line of the plane through the common point. Triangle-congruence arguments are then used to prove that the candidate spatial line makes a right angle with that arbitrary transversal.

For the formal reconstruction the significant classical features are:

- the target is universal over every line of the plane through the common point;
- the arbitrary transversal is part of the proposition's logical content, not merely a feature of the diagram;
- the proof is synthetic and based on incidence, right angles, equal segments, and triangle congruence.

A.3 Proposition XI.5

Proposition XI.5 says that if one straight line is perpendicular at a common point to three straight lines meeting there, then those three straight lines lie in one plane.

Euclid's contradiction argument uses the plane through two of the three candidate coplanar lines and another plane through the common perpendicular and the remaining line. The intersection line of the two planes produces, inside one plane, two distinct lines through the same point both perpendicular to the common perpendicular.

The local classical dependency is therefore governed by the preceding spatial propositions: the intersection-of-planes configuration and Proposition XI.4 supply the contradiction. The current Lean proof organizes this material differently, but preserves the same spatial core: perpendicularity to a plane forces the required intersection line to be perpendicular to the common normal.

A.4 Proposition XI.6

Heath gives the statement:

> If two straight lines be at right angles to the same plane, the straight lines will be parallel.

Let AB and CD be perpendicular to the reference plane at B and D. Euclid joins BD, draws DE in the reference plane perpendicular to BD, makes

$$DE = AB,$$

and joins BE, AE, and AD.

Since AB is perpendicular to the reference plane,

$$\angle ABD, \qquad \angle ABE$$

are right. Since CD is also perpendicular to the same plane, the corresponding lines of the plane through D are perpendicular to CD.

The first triangle comparison uses

$$AB = DE, \qquad BD = DB,$$

and equality of the included right angles. Proposition I.4 gives

$$AD = BE.$$

The second comparison uses

$$DE = BA, \qquad EA = AE, \qquad DA = BE.$$

Proposition I.8 therefore gives

$$\angle EDA = \angle ABE.$$

Hence $\angle EDA$ is right and

$$ED \perp DA.$$

At D, the line ED is now perpendicular to

$$DB, \qquad DA, \qquad DC.$$

Proposition XI.5 places these three lines in one plane. Proposition XI.2 supplies the corresponding triangle-plane incidence, bringing AB into that same plane. The two right angles made with the transversal BD then give the final parallelism by Proposition I.28.

The local classical DAG can be recorded as

$$\begin{gathered}\text{construct } DE \perp BD, \quad DE = AB \\ \Downarrow \\ \text{I.4} \\ \Downarrow \\ AD = BE \\ \Downarrow \\ \text{I.8} \\ \Downarrow \\ ED \perp DA \\ \Downarrow \\ \text{XI.5} \\ \Downarrow \\ DB, DA, DC \text{ coplanar} \\ \Downarrow \\ \text{XI.2} \\ \Downarrow \\ AB, BD, DC \text{ coplanar} \\ \Downarrow \\ \text{I.28} \\ \Downarrow \\ AB \parallel CD.\end{gathered}$$

A.5 Comparison with the Lean reconstruction at XI.6

The Lean proof preserves the central metric sequence of the classical argument:

$$\text{auxiliary } DE \longrightarrow \text{SAS} \longrightarrow \text{SSS} \longrightarrow \text{XI.5}.$$

The formal expansion is concentrated in the data suppressed by the classical diagram:

- each planar construction is performed in an explicit induced plane geometry;
- cross-plane segment and angle comparisons use the spatial congruence layer;
- noncollinearity and point-in-plane obligations are explicit;
- the common plane of the two final normals is produced as a witness;
- spatial parallelism records both coplanarity and disjointness, so skew lines are excluded;
- the last planar disjointness proof uses the established neutral alternate-angle infrastructure rather than a direct call to the public I.28 wrapper.

The last point is a dependency-graph difference, not a change in the geometric target. Euclid closes the configuration by I.28; the current Lean source imports the Proposition I.27 module and invokes the lower alternate-angle criterion in the induced common plane.

A.6 Checkpoint XI.6

Through XI.6 the reconstruction remains synthetic. No coordinate, vector, scalar-product, trigonometric, or numerical-angle representation is used in the proposition proofs.

The source comparison also isolates the new structural point of XI.6: in three-space the proof of parallelism has to establish coplanarity before planar nonintersection can be invoked. Euclid obtains this by XI.5 and XI.2. The Lean theorem records the same distinction directly in the target relation `HilbertSpaceLinesParallel`, which consists of an explicit common plane together with disjointness of the two ambient carriers.

A.7 Proposition XI.7

Heath gives the statement:

> If two straight lines be parallel and points be taken at random on each of them, the straight line joining the points is in the same plane with the parallel straight lines.

Let AB and CD be parallel straight lines and let $E \in AB$, $F \in CD$. Euclid supposes that the joining straight line EF does not lie in the plane of the parallels but in a more elevated plane. A plane is drawn through the alleged elevated join.

By Proposition XI.3, the intersection of that plane with the plane of the parallels is a straight line. Since E and F belong to both planes, the common section passes through E and F. Euclid then obtains the impossible situation described in the classical text as two straight lines enclosing an area.

Thus the explicit local Book XI dependency is

$$\text{XI.3} \longrightarrow \text{XI.7}.$$

A.8 Comparison with the Lean reconstruction at XI.7

The production Lean theorem has a shorter local dependency graph. Spatial parallelism is represented by

```
HilbertSpaceLinesParallel Geo l m
```

which already supplies an explicit common carrier plane for the two parallel lines, together with disjointness of their ambient carriers.

If E and F are points of the two parallel lines, their membership in that common plane follows immediately. Disjointness implies $E \neq F$, so the ordinary incidence structure constructs the joining line. The final containment is then obtained from

```
HilbertSpaceIncidence.line_in_plane
```

which is the formal spatial-incidence clause corresponding to Hilbert I.6.

Consequently the Lean theorem does not invoke Proposition XI.3. This is a difference in foundational organization, not in the geometric target: Euclid derives the line-in-plane conclusion by contradiction from a theorem about the intersection of two planes, whereas the Hilbert spatial incidence system contains the required line-in-plane closure directly.

A.9 Checkpoint XI.7

Through XI.7 the reconstruction remains synthetic. Proposition XI.7 is particularly economical: no order, congruence, right-angle, perpendicularity, induced-plane, coordinate, vector, scalar-product, trigonometric, or numerical-angle machinery enters its proof.

The classical and formal dependency graphs differ sharply at this point:

Euclid/Heath	Lean/Hilbert interface
XI.3	spatial parallelism witness
$\Downarrow$	$\Downarrow$
two-plane section	E, F in one plane
$\Downarrow$	$\Downarrow$
reductio	Hilbert I.6
$\Downarrow$	$\Downarrow$
XI.7	XI.7.

Appendix B

Hilbert: Spatial Axiomatic Architecture

B.1 Scope

This appendix records the Hilbert-side axiomatic audit for Euclid Book XI through Proposition XI.7. The relevant question is which classes of incidence, order, and congruence are used by the formal theorem and whether the Euclidean parallel axiom or a continuity principle enters.

B.2 Spatial layers in the formal development

The project separates the ambient spatial structure into

```
HilbertSpacePrimitive
HilbertSpaceIncidence
HilbertSpaceOrder
HilbertSpaceCongruence
```

and obtains ordinary planar Hilbert geometry on each fixed plane through

```
PlanePoint
PlaneLine
PlaneGeo
```

together with explicit bridges for incidence, betweenness, congruence, same rays, angles, and right angles.

No global

```
HilbertCongruence Geo
HilbertOrder Geo
```

instances are installed on the ambient three-space. Planar order and planar angle arguments are carried out in a specified `PlaneGeo`; spatial congruence transports data between the required plane slices.

B.3 Perpendicularity predicates

For line–line perpendicularity the general interface is

```
HilbertLinesPerpendicularAt
```

and includes an explicit noncollinearity witness for the two arms of the right angle. Hence line distinctness is derived by

```
hilbert_linesPerpendicularAt_ne
```

rather than supplied separately.

Line–plane perpendicularity is represented by

```
HilbertLinePerpendicularPlaneAt
```

which requires the line to be perpendicular, at the common point, to every line of the plane through that point.

B.4 Axiomatic status of XI.4

The public theorem assumes the spatial incidence/order/congruence hierarchy but no

```
HilbertEuclideanPlane Geo
```

instance. Thus the Euclidean parallel axiom, Hilbert Group IV, is not a hypothesis of XI.4 in the current formal hierarchy.

The axiom audit reported by Lean is

```
'Geometry.euclid_proposition_11_4' depends on axioms:
[propext,
 Classical.choice,
 Geometry.bookZero_nullSegment1,
 Quot.sound]
```

The inherited `bookZero_nullSegment1` dependency is retained at the current checkpoint. No proposition-specific geometric axiom is introduced by XI.4.

B.5 Checkpoint XI.4

The formal proof closes in the spatial incidence/order/congruence layer. No Group IV assumption and no new continuity assumption occur in the public theorem.

B.6 Proposition XI.7 and Hilbert I.6

Proposition XI.7 has a direct counterpart in Hilbert's spatial incidence axioms.

Hilbert's Group I, Axiom I.6 states that if two points A, B of a line a lie in a plane α, then every point of a lies in α. Hilbert immediately identifies this with the assertion that the whole line lies in the plane.

The same source explicitly distinguishes Axioms I.1–I.3 as the plane axioms of Group I and Axioms I.4–I.8 as the space axioms of Group I. Thus I.6 belongs exactly to the spatial incidence layer.

The Lean interface represents this clause by

```
HilbertSpaceIncidence.line_in_plane
```

In Proposition XI.7 the hypothesis

```
HilbertSpaceLinesParallel Geo l m
```

supplies a plane σ containing both parallel carriers. Hence the chosen points $E \in l$ and $F \in m$ both lie in σ. Their distinctness follows from disjointness of l and m. After the line through E and F is constructed, `HilbertSpaceIncidence.line_in_plane` places the entire line in σ.

No order axiom, congruence axiom, Group IV parallel axiom, or continuity axiom is used in this proof. No induced `PlaneGeo` is required.

B.7 Axiomatic status of XI.7

The Lean audit command

```
#print axioms Geometry.euclid_proposition_11_7
```

reports

```
'Geometry.euclid_proposition_11_7' does not depend on any axioms
```

This statement concerns global Lean axiom constants. The spatial incidence structure remains an explicit theorem parameter, so the result should not be read as eliminating Hilbert I.6 from the mathematical hypotheses. Rather, the proof term introduces no additional global axiomatic dependency beyond its explicit parameters.

B.8 Checkpoint XI.7

At XI.7 the formal development reaches a pure spatial-incidence proposition. The decisive source clause is Hilbert I.6, one of the space axioms of Group I. Spatial order, spatial congruence, Group IV, and continuity do not enter the theorem.

The formal theorem therefore gives a particularly direct test of the intended 3D incidence interface: the project declaration `HilbertSpaceIncidence.line_in_plane` is not an ad hoc XI.7 helper but the explicit Lean representation of Hilbert I.6.

Appendix C

Borsuk–Szmielew: Axiomatic Layers and Spatial Geometry

C.1 Scope

This appendix records the Book XI audit against Karol Borsuk and Wanda Szmielew, *Foundations of Geometry*, through Proposition XI.7. Their primitive language includes points, lines, planes, betweenness, and equidistance. Their axioms are organized into incidence, order, congruence, continuity, and the Euclidean axiom.

C.2 Congruence results relevant to XI.4

The Book I audit already identifies the triangle-congruence results used in the XI.4 argument within the pre-Euclidean congruence layer. In particular:

- the SSS theorem corresponding to Euclid I.8 occurs as Borsuk–Szmielew Theorem 21;
- right-angle and perpendicular constructions belong to their synthetic congruence development and are not formulated by numerical angle measure.

The formal XI.4 proof likewise treats SAS/SSS and right-angle transport as congruence infrastructure rather than as consequences of the Euclidean parallel axiom.

C.3 Spatial geometry

Borsuk–Szmielew develop three-dimensional geometry with points, lines, planes, perpendicular lines, perpendicular projection, and later an absolute coordinate system in space. In that spatial development a line perpendicular to a plane is part of the geometric language before coordinates are introduced as a later representation.

At the current checkpoint no theorem number has been identified in the audited source whose statement is exactly Euclid XI.4. Accordingly the Lean theorem is not attributed here to a specific Borsuk–Szmielew theorem.

C.4 Axiomatic comparison

For XI.4 the comparison is:

$$\begin{gathered}\text{incidence and order in planes}\\ +\\ \text{synthetic segment and angle congruence}\\ +\\ \text{spatial incidence/congruence transport}\\ \Downarrow\\ \text{line–plane perpendicularity.}\end{gathered}$$

The formal proof does not use the Euclidean parallel axiom. This agrees with the placement of the triangle-congruence ingredients below the Euclidean layer in the Borsuk–Szmielew organization.

C.5 Checkpoint XI.4

The current audit records no exact Borsuk–Szmielew counterpart for XI.4 itself. Exact theorem-number correspondence should be added only if it is located in the source.

C.6 Proposition XI.7 and incidence Axiom I7

For XI.7 the Borsuk–Szmielew source contains an exact structural counterpart of the decisive Hilbert incidence clause.

Their Axiom I7 states that for a line L and a plane P, if there exist two distinct points a, b belonging both to L and to P, then

$$L \subset P.$$

This is exactly the line-in-plane principle used in the formal proof of XI.7. Borsuk–Szmielew also classify Axioms I1–I4 as plane axioms and Axioms I5–I9 as space axioms. Hence I7 belongs to the spatial incidence layer.

Their general organization is especially transparent here: the first group consists of incidence axioms concerning points, lines, and planes; order, congruence, continuity, and the Euclidean axiom are introduced only in later groups. Proposition XI.7 therefore lies strictly below all of those stronger layers.

The correspondence with the Lean proof is

$$\begin{gathered}E, F \in \sigma, \qquad E \neq F\\ \Downarrow\\ \text{line } n \text{ through } E, F\\ \Downarrow\\ \text{Borsuk–Szmielew I7 / Hilbert I.6}\\ \Downarrow\\ n \subset \sigma.\end{gathered}$$

No Borsuk–Szmielew theorem number is needed here: the relevant source item is the incidence axiom itself.

C.7 Checkpoint XI.7

At XI.7 the comparison becomes sharper than at XI.4. The source contains an explicit spatial-incidence axiom whose content matches the decisive formal step exactly.

Thus XI.7 requires only incidence structure in the Borsuk–Szmielew hierarchy. It does not require their axioms of order, congruence, continuity, or the Euclidean axiom. This agrees with the Lean theorem, which assumes only the ambient incidence classes and whose axiom audit reports no additional global axiom constants.

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[4] Henry George Forder, *Foundations of Euclidean Geometry*, Cambridge University Press, Cambridge, 1927.